



A micro-based approach to evaluate the effect of water supply on health in Uganda



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ABSTRACT

This paper proposes a micro-based approach to investigate the impact of improved water provision on individual health outcomes in rural Uganda. We merge household and individual panel datasets with sub-county level administrative data on water supply projects. Our approach allows us to estimate fixed-effect panel data models which use temporal and spatial variation at the sub-county level as identifying variation. We find evidence that the installation of improved water sources leads to higher reported improved water usage, and shorter water collection times. However, increasing the sub-county rate of improved water sources per capita does not seem to lead to a statistically significant effect in the likelihood of individuals suffering from symptoms of illness associated with inadequate water supply. We argue that our micro-based approach provides a cost-effective means of evaluating development projects. The approach is scalable, i.e. it can be applied to other settings and countries.

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1. Introduction

In 2015, an estimated 844 million people worldwide lacked access to basic safe drinking water services (WHO & UNICEF, 2017). The situation is especially severe in sub-Saharan Africa, with people living in this region accounting for almost half of the world's population who do not have access to an improved¹ water source (WHO, 2015). Achieving “universal and equitable access to safe and affordable drinking water for all by 2030” is among the United Nation's Sustainable Development Goals. Governments and international donors support campaigns and water infrastructure projects to increase and improve water supply.² Such projects offer

potential welfare improvements, including the reduction of water-borne diseases and the reduction in water collection times by women or children.

Water supply interventions have been extensively studied (see Wolf et al. (2018) for a review). Apart from methodological and implementation challenges, the cost of randomisation has been a significant hindrance in conducting large-scale impact evaluation studies of water supply infrastructure projects. With the notable exceptions of Devoto et al. (2020) and Opryszko et al. (2010), studies on water supply interventions use data which result from potentially contentious choices regarding control groups, use comparatively small sample sizes, or are limited in geographical and temporal scope. These challenges may affect the external validity of the literature's findings.

We suggest a micro-based approach to evaluate the impact of water supply interventions on the use of improved water and the health outcomes of such use. We match an existing water supply programme data with geo-coded household panel surveys. This approach allows us to account for fixed-effects in our analyses so that we can exploit the spatial and temporal variation in programme intensity to control for unobserved differences across households and geographical regions. In total, we investigate over

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¹ Improved drinking water sources are defined as those that “by nature of their construction, are likely to be protected from outside contamination, in particular from contamination with faecal matter” (WHO/UNICEF Joint Monitoring Programme, 2015b). Examples are deep boreholes, public taps, protected springs, and shallow wells.

² Adding more water points can be seen as one way to improve water supply.

1,000 households in 318 sub-counties in Uganda from 2009 to 2012.

There are at least five main advantages to our micro-based approach with the individual fixed-effects that also apply for settings outside of Uganda: (1) it allows us to study the impact of water supply investments on a large, often nationally-representative sample of households over a longer period, rather than limiting the analysis to a single cross-section; (2) the panel structure allows controlling for *unobservable* time-invariant factors³; (3) given the richness of various household surveys, it is possible to control for many confounding time-variant factors; (4) it allows for reasonable identification in cases where randomised control trials are not feasible or are too costly; (5) due to their low frequency and the variety of topics addressed in household surveys, estimates of treatment effects are less likely to suffer from a *Hawthorne effect* (i.e., a bias that occurs if the act of being surveyed itself causes behavioural change [Zwane et al., 2011](#)).

Importantly, using an already collected national household panels makes our methodology much less costly for analysts and researchers compared to a randomised study of a similar sample size. Moreover, researchers can rely on already existing datasets which can be merged on geo-codes. Such studies can be conducted within a larger geographical area and across urban and rural areas, as is the case in this article. Our evaluation approach is informed by [Gertler and Molyneaux \(1994\)](#), [Pitt, Rosenzweig, and Gibbons \(1993\)](#), [Rosenzweig & Wolpin \(1986\)](#) and, more recently, [Zhang \(2012\)](#). We also relate to the literature which uses difference-in-difference estimators to study the impact of improved water provision (e.g. [Giliani et al. 2009](#) and [Gross et al. 2018](#)).

Aside from the methodological innovations, we provide evidence on the importance of proximity to improved water sources, in addition to the common socio-cultural factors in the literature ([Wasonga, Okowa, & Kioli, 2016](#); [UNESCO, 2002](#)). Once controlling for household (or individual) fixed-effects, it is not, *a priori*, clear that the provision of water sources in the proximity of the household would automatically increase improved water usage, decrease disease incidence, and reduce the amount of time spent on fetching water. Employing our setting, we find some evidence that an improved water supply increases the likelihood of improved water usage, in that it may reduce the presence of water-related disease symptoms and may reduce water collection times.

The remainder of this paper is structured as follows: Section 2 provides a review of the literature; we discuss our data and methodology in Section 3; Section 4 provides results for the impact of improved water provision on usage, health outcomes and water collection times; and Section 5 offers concluding remarks.

2. Related literature

Following the early works of [Esrey, Potash, Roberts, & Shiff \(1991\)](#), [Esrey, Feachem, & Hughes \(1985\)](#), and [Blum and Feachem \(1983\)](#), researchers have intensively studied the health impact of water and sanitation interventions (e.g., [Aiello, Coulborn, Perez, & Larson, 2008](#); [Amrose, Burt, & Ray, 2015](#); [Cairncross et al., 2010](#); [Clasen, Schmidt, Rabie, Roberts, & Cairncross, 2007](#); [Ejemot-Nwadiaro, Ehiri, Arikpo, Meremikwu, & Critchley, 2015](#); [Loevinsohn et al., 2015](#); [Norman, Pedley, & Takkouche, 2010](#); [Taylor, Kahawita, Cairncross, & Ensink, 2015](#)). Published studies

generally tend to find certain health effects of water projects in low- and middle-income countries.

Many researchers have pointed out the methodological challenges that are associated with the approaches employed in numerous studies in this field (e.g., [Dupas & Miguel, 2017](#); [Blum & Feachem, 1983](#); [Fan & Mahal, 2011](#); [Strunz et al., 2014](#)). Often, the use of cross-sectional data does not clearly identify the effects of water projects, and relevant control groups are hard to find (e.g., [Majuru, Michael Mokoena, Jagals, & Hunter, 2011](#); [Nanan, White, Azam, Afsar, & Hozhabri, 2003](#); [Tonglet, Isu, Mpese, Dramaix, & Hennart, 1992](#)). Moreover, some of the studies offer limited scope due to their small sample sizes, as large samples are expensive to gather (e.g., [Aziz et al., 1990](#); [Blum et al., 1990](#); [Gasana, Morin, Ndikuyezze, & Kamoso, 2002](#); [Hasan et al., 1989](#); [Huttly et al., 1990](#); [Madrigal, Alpizar, & Schlüter, 2011](#); [Majuru, Michael Mokoena, Jagals, & Hunter, 2011](#); [Ryder et al., 1985](#); [Tonglet, Isu, Mpese, Dramaix, & Hennart, 1992](#); [Zeng-Sui et al., 1990](#)). Similarly, geographical coverage may be limited, which can lead to concerns regarding external validity.

Observational and randomised evaluations may be associated with the *Hawthorne effect* (i.e., study participants may alter their behaviour due to their awareness of being observed). Indeed, [Zwane et al. \(2011\)](#) show that Kenyan households that are frequently surveyed about their children's health status behave differently.⁴ Frequent surveys may affect the estimated treatment effect of ongoing water quality interventions ([Kremer et al., 2011](#)). The *Hawthorne effect* has important implications because frequent data collection is typical in epidemiology and, consequently, the effect may bias evaluations of water supply intervention. A potential solution is to collect more data, but passively, through administrative data, or 'pure control' groups (see [Dupas & Miguel, 2017](#)). Evidently, the use of administrative sub-county water supply partially solves this problem, but the repeated interviews in the panel could also lead to the *Hawthorne effect*.⁵

The literature has also applied econometric techniques such as propensity score matching (PSM) or similar methods on cross-sectional, nationally-representative household survey data ([Begum, Ahmed, & Sen, 2011](#); [Capuno, Tan, & Fabella, 2015](#); [Fan & Mahal, 2011](#); [Fink, Günther, & Hill, 2011](#); [Jalan & Ravallion, 2003](#); [Khanna, 2008](#); [Roushdy, Sieverding, & Radwan, 2013](#)). Such approaches have the advantages of using larger sample sizes, being less likely to suffer from bias from the *Hawthorne effect*, and employing noteworthy identification strategies.⁶ PSM is most useful when only observed characteristics are believed to affect treatment assignment and outcomes. This is unlikely to be the case with water supply projects. As [Pitt, Rosenzweig, & Gibbons \(1993\)](#) argue, coverage and timing of water-related interventions may be responsive to unobserved characteristics of the targeted populations, which may influence outcomes of interest and introduce bias. For example, to maximise their impact, water supply interventions may be targeted at areas with low coverage of basic health care while the local availability of health care is likely also to affect households' health outcomes. PSM methods that do not account for the location-specific provision of health care will lead to biased estimates of the health effects of water supply interventions.

⁴ The authors identify a *Hawthorne effect* by showing that more frequent health surveys lead to statistically significant effects on the usage of chlorine disinfectant, reported child diarrhoea, and measured water quality.

⁵ It may be sensible to expect that a *Hawthorne effect* is less relevant for general surveys on a large number of topics than for specific experiments that focus on one topic only (e.g. after an educational experiment participants may want to show to the experimenters next time that they have learned something).

⁶ In the case of PSM, the key assumption is *conditional independence* (i.e., given a set of observable covariates that are not affected by treatment, potential outcomes are independent of treatment assignment ([Gertler, Martinez, Premand, Rawlings, and Vermeersch \(2016\)](#))).

³ Many of the nationally representative surveys capture information on water use with longitudinal but not panel structures. However, household and individual panels are becoming increasingly popular in Africa and other developing countries through the World Bank's Living Standards Measurement Survey and others (see <https://microdata.worldbank.org/index.php/catalog/lsms> for some of the latest panel survey available). Employing data from the Demographic Health Survey allows to look at the level of individuals too while accounting for the enumeration area.

An alternative is to employ fixed-effects methods to study programmes by governments in developing countries. Pitt, Rosenzweig and Gibbons (1993), Rosenzweig and Wolpin (1986), & Gertler and Molyneaux (1994), and Gertler & Molyneaux, 1994 have employed this technique in different studies including planning, health, and schooling programmes in the Philippines and Indonesia.⁷ Duflo (2001) used a similar approach to evaluate schooling and labour market consequences of school construction. We contribute to this literature by merging available geo-coded household panel data with geo-coded data on water supply projects in Uganda. Few studies are closely related to what we do in this paper. Zhang (2012) reconstructs programme data from a household panel survey to evaluate the impact of introducing village-level access to water plants on health in rural China. Using a difference-in-difference estimation while controlling for the province- and year-fixed effects, Dolan et al. (2019) show that the long-lasting insecticide nets distribution programme leads to a decline in under-5 mortality risk among children in the Democratic Republic of Congo.⁸

3. Data and identification strategy

3.1. Data and merging of geo-coded households to water projects

Following the Uganda water sector reform of the mid-1990s, the government intended to provide access to safe water for 80 percent of the population by 2020 (Ministry of Water & Lands/Environment, 1999; National Planning Authority, 2015). From 1990 to 2015, access to at least basic drinking water increased from 24 to 32 percent between 2000 and 2015 according to the Joint Monitoring Programme (WHO & UNICEF, 2017).

We draw on panel data from two sources and merge them based on geographical location of households and water supply units. We rely on (1) the Uganda Water Supply Atlas (UWSA) for data on improved water sources; and (2) the Uganda National Panel Survey (UNPS) for data on household and individual-level outcomes.

The UWSA is published by the Ugandan Ministry of Water and Environment through its Directorate of Water Development. It gives an overview of the water supply situation in Uganda. The online version of the Atlas contains data on location, type, functionality, etc. of all improved water sources in Uganda according to the Ministry (i.e., including sources developed by NGOs).⁹ We directly accessed the sub-county information and the type of improved water from the website. However, data on the year of construction was not available, and we generated this variable from the various annual sub-county level reports for the financial years 2009/10 to 2015/16 that present aggregated data from the UWSA. For each year and sub-county, we obtained data on the number and type of improved water sources, their functionality, and population figures. These reports are accessible via an internal interface of the UWSA website. Since missing data was not coded consistently, we assumed missing data for rural sub-counties which consistently report zero functional or non-functional improved water sources (2.5 percent of all rural sub-county annual observations).

We obtain household and individual-level information from the UNPS. This is a multi-purpose data set which contains detailed information on households and their members. We use three

rounds of data (from 2009/2010 to 2011/2012) which are available at the World Bank Living Standards Measurements Survey's website. Aside from the socio-economic variables, the data contain geographic information and geo-coordinates of the sampled households. This enabled us to match the households to the water supply data based on the sub-county and year of the survey. We merge the data sets based on the district, sub-county and year of data collection. This means that the water supply variable varies within samples in the same round depending on the year of data collection. We merged household for three waves of the UNPS to the water supply data provided by the UWSA. As the same households were observed several times, our approach allows us to account for fixed-effects for households or individuals.

Overall, about 71 percent of households reported that they drank from an improved water source in 2009/2010 and this increased to about 75 percent in the 2011/2012 sample.

3.2. Variables and measurement

In our analysis, we employ three main dependent variables and account for the updated definitions following the Joint Monitoring Program for Water Supply, Sanitation and Hygiene (JMP). Our variables are based on reported data in the UNPS.

- (1) The first dependent variable reflects the household's reported usage of improved water as the primary source of drinking water (*WaterUsage*). This variable is measured with a question in the UNPS that asks, "What is the main source of water for drinking for your household?" We define a household as using an improved water source if its main source of drinking water is a private connection to a pipeline, public taps, borehole, protected well/spring, vendor/-tanker truck, gravity flow scheme or rainwater (see WHO & UNICEF, 2017). All other responses are classified as non-improved. Thus, we generate a binary variable whose value is one if the household used any of those sources and zero if otherwise. Moreover, we generate an additional variable following the new definition of improved water by the JMP (WHO & UNICEF, 2017). For this variable we take into account the water quality and total water collection time (including queuing time). This additional variable has five ordered outcomes resulting in a *Safe Water Ladder*: (5 = Safely managed) – the household uses improved water which is located in the dwelling or residence; (4 = Basic) – the source of improved water is not more than 30 min for a round trip including queuing; (3 = limited) – the household uses improved water but collection time exceeds 30 min; (2 = unimproved) – drinking water is from unprotected dug well or spring; (1 = surface water) – when drinking water is directly from river dam, lake, pond, stream or irrigation (WHO & UNICEF, 2017).
- (2) To examine the effect of improved water provision on health, we employ a dichotomous variable which indicates whether an individual reported to be suffering from illness related to acute or chronic diarrhoea¹⁰ and vomiting (*ill*) during the last 30 days. We select these self-reported conditions because they represent symptoms of the consumption of contaminated water. Our second dependent variable is at the individual level. Its value is one if a member of the household suffered from acute or chronic diarrhoea and vomiting and

⁷ Pitt, Rosenzweig, & Gibbons, 1993 describe the comparative advantage of using panel instead of cross-sectional data in detail.

⁸ Combining data from the Demographic Health Survey from several countries and geo-coded aid datasets on World Bank and Chinese development finance, Cruzatti, Dreher, & Matzat, 2019 show that Chinese aid and concessional World Bank aid increase infant mortality. Baetschmann et al., 2020 explore geo-coded World Bank aid and merge it with data from the Demographic Health Survey.

⁹ The data set has been used, among others, by Foster (2013) in a quantitative study to investigate predictors of the functionality of community-managed hand pumps.

¹⁰ The WHO (2017) defines diarrhoea as the passage of three or more loose or liquid stools per day (or more frequent passage than is normal for the individual). We rely on a self-reported variable of whether the individual suffered from acute or chronic diarrhoea, since this is the only information provided in the survey instrument.

zero if otherwise. A limitation of this measure of diarrhoea incidence is the likelihood of underreporting (Feikin et al., 2010; Overbey, Schwab, & Exum, 2019; Schmidt et al., 2011).

- (3) In keeping with the development literature on water and household time use (Boone, Glick, & Sahn, 2011; Gross, Günther, & Schipper, 2018), we provide analyses of the effect of improved water provision on time spent on water collection.¹¹ Physical access has been cited as one of the main problems related to water. As has commonly been done in the existing literature, we use the total time spent on water collection by the household (*Time*) as a dependent variable. An increase in the installed capacity per capita should ultimately reduce the amount of time a household spends fetching water. Again, we use the self-reported¹² time (minutes) spent on water collection for household use in the last seven days. Besides, we separately analyse the effect of improved water supply on reported total walking and waiting time at the water source, and the geographical distance to the water source.

As our main independent variable, we measure the provision of improved water by the improved water supply capacity per capita (*WaterSupply*). The improved water supply rate was calculated by multiplying each improved water source in the sub-county with the number of people it was capable of serving.¹³ We then divide the total number of people that could be served in the sub-county by the population size to obtain the *improved water supply rate*. Our measure follows the methodology of the Uganda Water Supply Atlas (see MoWE, 2017). This variable takes into account the fact that different types of water projects may have different capacities. It is important mentioning that the improved water supply rate does not measure actual access to water, but describes the installed improved water supply capacity per capita that takes into consideration the different capacities of the various water projects (i.e., if improved water sources are not close to citizens, we still account for them when calculating the improved water supply rate). Finally, for the classification of improved community water sources, the definition of the UNICEF/WHO JMP was applied. At baseline, the average improved water supply rate was 0.59 (About 59 percent of people potentially have access to improved water). Yearly changes in the improved water supply rate can be negative if water sources break down or the population grows faster than the number of people that can be served by improved community water sources.

The use of a multi-purpose household dataset such as the Uganda National Panel Survey (UNPS) allows us to account for various control variables in our empirical setting. These include whether the household head has been to school, household size, household expenditure, etc. All variables and descriptive statistics for the years of analysis are presented in Table A1 in the Appendix.

3.3. Identification strategy

Our empirical identification strategy is straightforward. The improved water supply capacity per capita (*WaterSupply*) is measured in the sub-county where the household was located at the time of the survey. We observe the households several times in the panel, and household-specific (as well as community-

specific) control variables are available. For time-invariant unobservables, we employ household fixed-effects (or individual fixed-effects where appropriate) and account for year fixed-effects. Thus, we estimate the following three specifications for each of the dependent variables, *WaterUsage*, *Ill*, and *Time*:

$$WaterUsage_{ht} = \psi_1 WaterSupply_{ht} + \beta_h + C_{ht}\psi_2 + \phi_t + \epsilon_{ht} \quad (1)$$

$$Time_{ht} = \pi_1 WaterSupply_{ht} + \delta_h + C_{ht}\pi_2 + \phi_t + \xi_{ht} \quad (2)$$

$$Ill_{it} = \theta_1 WaterSupply_{it} + \gamma_i + C_{it}\theta_2 + \phi_t + \eta_{it} \quad (3)$$

WaterUsage is the indicator variable measuring whether the household *h* essentially drinks from an improved water source or not. *Ill* is the indicator variable that indicates whether an individual *i* in the household suffered from a water-related illness or not. *Time* is the total household time spent on water collection. Since the dependent variables of (1) and (3) are dummies, we estimate them using a Logit estimator (linear probability models yield qualitatively similar results).¹⁴ Eq. (2) is estimated with OLS. β_h , γ_i and δ_h represent household fixed-effects (Eqs. (1) and (2)) and individual fixed-effects (Eq. (3)). *C* represents a design matrix which includes household and individual characteristics when appropriate (see Table 1) as well as characteristics of the community where the household is located, such as access to a road and health facility. ϕ accounts for time fixed-effects.

Our main independent variable of interest is *WaterSupply*. The coefficients, ψ_1 , θ_1 and π_1 capture the impact of the improved water supply capacity per capita. As we account for household (or individual) fixed-effects, we can investigate how a change in *WaterSupply* affects the same household (or the same individual) over time, i.e. we investigate whether an increase in the improved water supply capacity per capita leads to increased improved water usage, less water-related illness, and lower water collection times.

Household (or individual) fixed-effects go a substantial way towards reducing endogeneity issues in comparison to the existing literature. Households in sub-counties with low levels of improved water coverage may vary from those in sub-counties with high levels of coverage in many ways that also matter for the outcomes of interest. If these differences remain constant throughout the sample period, however, they are controlled for by the inclusion of fixed-effects. Similarly, the 'bottom-up' approach to local water development planning may fuel concerns about the endogeneity of infrastructure placement (e.g., some communities develop faster than others and request or receive more improved water sources).¹⁵ Again, fixed-effects account for pre-existing infrastructure projects. The UNPS also allows us to account for time-variant characteristics of households so that we can further alleviate omitted variable bias. Finally, our data permits us to account for infrastructure developments over the sample period that are related to changes in roads and health facilities, which are included in the design matrix *C*.¹⁶ All standard errors are clustered at the sub-county level, which helps to deal with the potential autocorrelation problem (Bertrand, Duflo, & Mullainathan, 2004).

¹⁴ When using the Reported Safe Water Ladder as a dependent variable, we employ an ordered logit model.

¹⁵ Local characteristics may also affect health outcomes, and thus, if not controlled for, bias estimates of the effects of improved water sources on health outcomes.

¹⁶ There is the possibility for local politicians to influence the allocation of water sources and other infrastructure (Quin et al., 2011). Data from the Ugandan Water Supply Atlas provides some evidence that, for the sub-counties during the sample period, allocation regulations were generally followed. The data suggest that rural sub-counties that had a safe water coverage below the national average (and should, therefore, have qualified for more resources) saw an increase in the rate of improved water sources per capita in the following year.

¹¹ We do so even if time is reflected in the JMP ladder to make our results comparable to previous literature and as time continues to be used as a direct outcome variable.

¹² We acknowledge that self-reported collection time may be overstated (Pearson (2016); Ho, Russel, and Davis (2014)). If overstatement is not more likely for cases where supply of improved water has increased, overstatement should not affect our results.

¹³ Table A2 in the appendix provides information for the various water sources (see MoWE, (2017)).

Table 1
Effects of improved water supply on reported improved water usage.

	(1)	(2)	(3)	(4)
<i>Panel A: Reported Improved Water Usage</i>				
	Logit-RE	Logit-RE	Logit-FE	Logit-FE
Improved water supply capacity per capita	0.128* (0.069)	0.145** (0.073)	0.373* (0.212)	0.502** (0.238)
Household and community time-variant controls	No	Yes	No	Yes
Year fixed-effects	No	Yes	No	Yes
Household fixed-effects	No	No	Yes	Yes
Observations	4800	4800	1028	1028
Wald χ^2	5.418 [0.020]	52.969 [0.000]	3.109 [0.078]	38.055 [0.000]
Pseudo R ²			0.004	0.055
<i>Panel B: JMP Safe Water Ladder – Ordered Logit Models</i>				
	Ologit-RE	Ologit-RE	Ologit-FE	Ologit-FE
Improved water supply capacity per capita	0.746** (0.332)	0.742** (0.346)	0.954 (0.601)	1.665** (0.688)
Household and community time-variant controls	No	Yes	No	Yes
Year fixed-effects	No	Yes	No	Yes
Household fixed-effects	No	No	Yes	Yes
Observations	4800	4800	3950	3950

Notes: In panel A, logit models are estimated, and marginal effects are reported. In panel B, ordered logit models are estimated for the dependent variable JMP Safe Water Ladder (see text for description) and coefficients are reported. Household and community time-variant controls include all household level controls shown in Table A1 in the appendix. Estimated standard errors in parentheses are clustered at the sub-county level. * $p < .1$, ** $p < .05$, *** $p < .01$. [#] denotes p-value of statistic. Sample differences between Logit-RE and Logit-FE are due to the strict reliance of the conditional fixed effect logit model on variations in the outcome variable.

Our approach comes with challenges: (1) parameter estimates of treatment effects may still be affected by other *time-variant* unobserved factors that are correlated with treatment, such as heterogeneity in sub-county economic development; (2) the reliance on secondary data sets limits to our choices, i.e. we can only deal and approach research questions that can be addressed with the questions in the respective datasets; (3) since the water programme data are drawn from administrative records, the quality of the final data and sample depend on the completeness of the information provided. Bias from measurement error may be increased by transformations to estimate the model resulting in less efficient estimates (Angrist & Krueger, 1999); (4) While it is possible to investigate medium to long term effects of programmes, any implementation is naturally restricted by the length of the available panel datasets. It has to be highlighted that in the medium to long term those benefits attributed to programmes can dissipate and even disappear. Keeping these challenges in mind, we believe that our micro-based approach with fixed-effects offers a practical method for evaluating development programmes ex-post. This will be useful for researchers since, amongst other institutions, the World Bank, in collaboration with various statistical offices, collects panel data in several countries.

4. The impact of improved water supply

4.1. Effect on the usage of improved water

Table 1 provides results for the impact of improved water supply capacity per capita on a household's reported source of drinking water in Panel A. Models (1) and (2) provide random effect models and (3) and (4) present conditional logit estimates with household fixed-effects. The reduced sample sizes of models (3) and (4) is explained by the implementation of the Logit-FE estimation through the software (Stata) which requires variation in the dependent variable over time. Hence, households that did not change their main source of drinking water are not used in these models.

In model (1), the household's reported use of improved drinking water is regressed only on the provision of improved water, without any additional control variables or household fixed-effects. In

model (2), other control variables, comprising household and community time-variant characteristics and time fixed-effects, are added. Model (3) estimates improved drinking water as a function of improved water provision and household fixed-effects, while all covariates are added to the household fixed-effects in the model (4). Thus, model (4) fully implements our identification strategy and is our preferred specification. Due to the inclusion of household fixed-effects the results of models (3) and (4) can be interpreted to show the average semi-elasticity of reported improved water usage with respect to a *change* in the improved water supply rate.

All the results of Table 1 show a positive effect of improved water supply on the probability of reported improved water usage. Quantitatively, model (4) indicates that the probability that a household drinks from an improved water source increases by about five percentage points if the improved water supply at the sub-county increases by 10 percent.

In Panel B, we combine the reported water collection time and the type of water to generate a five-outcome ordered categorical variable (JMP Safe Water Ladder). Following Baetschmann, Ballantyne, Staub, & Winkelmann, 2020; Baetschmann, Staub, & Winkelmann, 2015 we estimate fixed-effects ordered logit models to analyse the effect of improved water supply on the JMP Safe Water Ladder and present the raw coefficients. The results indicate that improved water supply at the sub-county level increases the chances that a household will move up on the JMP Safe Water Ladder. All coefficient estimates are positive and statistically significant.

4.2. Effect on health

Given that the use of improved water increases with its supply, we investigate whether improved water supply affects people's health (Eq. (3)). Table 2 presents the result for the dependent variable *Ill*, which captures water-related disease symptoms of individuals within the household in the last thirty days before the survey (e.g., diarrhoea). Our estimations are performed for individuals, and we use individual fixed-effects (instead of household fixed-effects).

Table 2 does not yield any statistically significant effects when we employ either random effects (models 1 and 2) or fixed-effects

Table 2
Impacts of improved water supply on water-related disease symptoms (diarrhoea).

	(1)	(2)	(3)	(4)
	Logit-RE	Logit-RE	Logit-FE	Logit-FE
Improved water supply capacity per capita	0.002 (0.005)	0.005 (0.005)	0.693 (0.693)	0.429 (0.613)
Individual, household and community controls	No	Yes	No	Yes
Year fixed-effects	No	Yes	No	Yes
Household fixed-effects	No	No	Yes	Yes
Observations	23,772	23,772	2595	2595
Wald χ^2	0.108 [0.709]	341.359 [0.000]	1.030 [0.318]	634.892 [0.000]
Pseudo R ²			0.001	0.069

Notes: Logit models are estimated, and marginal effects are reported. Individual and community time-variant controls include all individual level controls shown in Table A1 in the appendix. Estimated standard errors in parentheses are clustered at the sub-county level. * $p < .1$, ** $p < .05$, *** $p < .01$. [#] denotes p-value of statistics. Sample differences between Logit-RE and Logit-FE are due to the strict reliance of the conditional fixed effect logit model on variations in the outcome variable.

Table 3
Effect of improved water provision on the household's reported total water collection time and distance.

	(1)	(2)	(3)	(4)
<i>Panel A: Reported Total Water Collection Time</i>				
Improved water supply capacity per capita	-10.890*** (4.014)	-7.319* (3.808)	-3.705 (10.784)	-4.158 (9.103)
Household and community time-variant controls	No	Yes	No	Yes
Year fixed-effects	No	Yes	No	Yes
Household fixed-effects	No	No	Yes	Yes
Observations	4634	4633	4634	4633
R ²	0.006	0.071	0.006	0.038
<i>Panel B: Reported Distance to Water Source</i>				
Improved water supply capacity per capita	-0.443*** (0.156)	-0.394*** (0.143)	-1.480 (1.006)	-1.881* (1.069)
Household and community time-variant controls	No	Yes	No	Yes
Year fixed-effects	No	Yes	No	Yes
Household fixed-effects	No	No	Yes	Yes
Observations	4775	4775	4775	4775
R ²	0.002	0.009	0.002	0.003

Notes: OLS models are estimated. Household and community time-variant controls include all household level controls shown in Table A1 in the appendix. Estimated standard errors in parentheses are clustered at the sub-county level. * $p < .1$, ** $p < .05$, *** $p < .01$.

(models 3 and 4).¹⁷ This finding deserves further discussion: A potential *a priori* expectation was a negative and significant effect of improved water supply on water-related illness symptoms. One possible reason for the non-significance of improved water provision could be the fact that diarrhoeal symptoms may be associated with different kinds of sicknesses, some of which are not water-related. Another possible reason is that the incidence of diarrhoea may be misreported (Feikin et al., 2011; Overbey et al., 2019). We also included the JMP Sanitation Ladder as a control of household's sanitation practices. This variable was statistically significant in model 3.¹⁸ Some literature suggests that the supply of improved water alone may not change hygiene behaviour and prevent diarrhoea. More comprehensive water and sanitation interventions which include behavioural changes may be needed (Cumming et al., 2019). Thus, our statistically insignificant results regarding diarrhoea could be interpreted to be in line with the evidence that water supply infrastructure alone is unlikely to be sufficient to have substantial effects on the diarrhoeal disease burden. The proposed micro-based evaluation approach in this paper could be extended to focus on other, and potentially more effective interventions apart from improved water supply.

¹⁷ Table S3 in the online supplement also suggests that none of the different elements that are relevant for the indicator "improved water supply" such as protected springs, deep boreholes or standpipes individually affect water-related disease symptoms. The only exception is for yard taps which show a negative and statistically significant effect.

¹⁸ We thank an anonymous referee for pointing out to add this control variable.

4.3. Effect on water collection time

We investigated the impact of improved water supply on reported total water collection times and the reported distance to the next water source in Table 3.

The results show a consistent negative effect of improved water installation on the reported amount of time spent on water collections. The effect is statistically significant when estimated by a random effects model. Quantitatively, the results in columns (1) and (2) indicate that total water collection time decreases on average by about 7 to 10 min per unit increase in the improved water supply capacity per capita, which is a comparatively small effect. Once household fixed-effects are included, the coefficient of improved water supply remains negative but loses statistical significance. In an online supplement (Table S1) we show that the reported reduction in total water collection time may be due to a reduction in walking time while waiting time at the water source is not statistically affected by an increase in the improved water supply capacity.¹⁹

Panel B indicates that improved water supply indeed improves access to water by reducing the distance to the water source. Model 4 shows that the average reported distance to the water source decreases by nearly 1.9 km per unit increase in the improved water supply in the sub-county.

¹⁹ Moreover, the potential time savings tend to be mostly due to the protected springs (Table S2 in the online supplement).

Methodologically, the results in Table 3 also serve as a robustness check on the findings and conclusions in Tables 1 and 2, as additional water supply should not increase collection times and distances overall even if citizens choose to use other sources, i.e. citizens may always choose to use their “old” supply instead of the new sources. More importantly, the results may be interpreted to provide indirect evidence of the relevance of water supply on issues such as household time allocation (especially for women and children), women’s empowerment, and poverty reduction (see Ivens, 2008; Ray, 2007). In a rural African context, we would, for instance, expect a reduction in water collection time to translate into more time spent on studies/leisure for children, and more time spent on the farm or commercial activities for women (Jasper, Le, & Bartram, 2012; Koolwal & van de Walle, 2013).

5. Concluding remarks

A significant proportion of development funds are devoted to water, sanitation, and hygiene projects. Development funds need to satisfy competing needs, and therefore assessing the impacts of diverse projects is relevant. The literature has investigated potential impacts of improved water provision on health and household welfare. While much of the literature shows a positive impact from water provision, methodological challenges in the design of the evaluations or the lack of funds for large-scale randomised control trials affect the extent to which policymakers can rely on the results.

We suggest a micro-based and cost-effective approach which entails evaluating past development projects and employing newly merged datasets to investigate the impact of improved water supply on water usage, health, and collection times. We apply our approach to the case of Uganda. The empirical results indicated modest but positive effects of improved water provision on a household’s reported usage of this water. We do not find evidence that an increase in the coverage of improved water is sufficient to decrease water-related illness symptoms in household members. However, water collection times and distances tend to be reduced as the supply of improved water increases in the case of Uganda.

The major contribution of our study, in addition to empirical findings for Uganda, is our suggestion of a micro-based evaluation approach. It employs available datasets and merges them for impact evaluations (i.e., we merge administrative information on water supply with household socio-economic panel data). Methodologically, the merging of data allows us to employ household and individual fixed-effects when investigating potential impacts of development projects. Our approach helps to systematically reduce endogeneity issues. Socio-economic data is collected in many countries at regular intervals, and samples tend to be large

in comparison to single project related surveys. Moreover, a *Hawthorne effect* is less likely for socio-economic surveys. Our approach also allows us to evaluate a large number of past projects at the same time, and it is, thus, scalable. As new data need not be collected, it is highly cost-effective.

There are, however, relevant limitations as socio-economic data may not be detailed enough for specific research purposes. Moreover, we are not able to control for all relevant other health and infrastructure related-projects in the sub-county aside from what is reported in the Uganda Water Atlas, i.e. the approach suffers as most other approaches from potential omitted variable bias due to unobserved/unmeasured other projects. Nevertheless, our approach likely has higher external validity than is typically achieved with randomised control trials. In any case, it offers a viable complement to standard methods of programme evaluation in developing countries, which can be conducted at low cost using pre-existing administrative data and micro-surveys, years after development projects have occurred. Finally, as numerous researchers now geo-tag conducted interviews, it should be possible in the future to merge such data with geo-tagged data from the Word Bank or national administrative records on projects such that there is large scope for future research.

CRediT authorship contribution statement

Raymond Frempong: Formal analysis, Writing - review & editing. **Lucas Kitzmüller:** Methodology, Validation, Formal analysis, Writing - review & editing. **David Stadelmann:** Conceptualization, Methodology, Validation, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1

Variable description and summary statistics.

Variable	2009/10	2010/11	2011/12	Average
Panel A: Variables used to estimate equations (1) and (2) – Household level				
Improved water supply capacity per capita	0.589	0.597	0.591	0.592
Household JMP Improved drinking water (No = 0, Yes = 1)	0.707	0.723	0.749	0.727
JMP Safe Water Ladder	2.907	3.011	3.122	3.016
Reported total water collection time (minutes)	54.85	46.31	39.35	46.68
Reported total waiting time (minutes)	22.10	17.36	12.19	17.11
Reported total walking time (minutes)	32.79	28.95	27.17	29.59
Reported distance to water source (km)	0.950	0.809	0.593	0.779
Number of protected springs per 1000 capita	0.867	0.826	0.795	0.828
Number of shallow wells per 1000 capita	0.379	0.400	0.397	0.392
Number of deep boreholes per 1000 capita	0.695	0.769	0.757	0.741
Number of rainwater tanks per 1000 capita	0.598	0.613	0.626	0.613
Number of public stand pipes per 1000 capita	0.490	0.388	0.440	0.439
Number of house connections per 1000 capita	0.178	0.160	0.0299	0.120
Number of institutional connection per 1000 capita	0.072	0.081	0.037	0.063
Number of yard taps per 1000 capita	0.367	0.485	0.564	0.474
Number of people in the Household	6.672	7.268	7.820	7.266
Age of household head	47.26	46.80	47.22	47.10
The household head has been to school (No = 0, Yes = 1)	0.772	0.824	0.793	0.796
JMP sanitation ladder	1.666	1.635	1.612	1.637
Log of household expenditure	10.72	10.53	10.57	10.61
Number of rooms in the house	2.953	2.968	3.020	2.981
The community has access to road (No = 0, Yes = 1)	0.987	0.972	0.983	0.981
Community has health facility (No = 0, Yes = 1)	0.924	0.810	0.541	0.753
Panel B: Variables used to estimate equation (3) – individual level				
The individual suffered from diarrhea (No = 0, Yes = 1)	0.067	0.055	0.035	0.052
Age of individual	23.98	24.07	24.13	24.06
The person has been to school (No = 0, Yes = 1)	0.821	0.846	0.842	0.836
Age of household head	48.08	47.94	48.21	48.08
The household head has been to school (No = 0, Yes = 1)	0.791	0.844	0.816	0.817
Number of people in the Household	5.394	4.993	5.036	5.141
Log of household expenditure	10.70	10.50	10.52	10.57
JMP sanitation ladder	1.706	1.664	1.648	1.672
Number of rooms in the house	3.338	3.330	3.352	3.340
The community has access to road (No = 0, Yes = 1)	0.988	0.972	0.985	0.982
The community has a health facility (No = 0, Yes = 1)	0.921	0.814	0.523	0.747

Notes: Statistics (means) are unweighted. Variables for individuals are aggregated to the household level before being averaged.

Sources: Uganda National Panel Survey (2009–2012) and the Uganda Water Supply Atlas.

Table A2

Improved Community Water Source User Number Assumptions.

Water source type	Users
Protected Spring	200
Shallow Well	300
Deep Borehole	300
Kiosk / Public Tap	150
Rainwater Harvesting Tank < 10,000 l	3
Rainwater Harvesting Tank greater than 10,000 l	6
Institutional connection	100
Yard Tap	24

Source: MoWE (MoWE).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.worlddev.2021.105522>.

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